Deep learning for Aerodynamic predictions

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What is CFD?

CFD is a simulation method that solves complex physics equations(Navier-Stokes) to calculate how fluid flows over a surface.

- Standard Methods: Uses physics to simulate the airflow.
- Tools: XFOIL, OpenFOAM, ANSYS.
- Output: Calculates lift(Cl) and Drag (Cd) coefficients.

PROBLEM

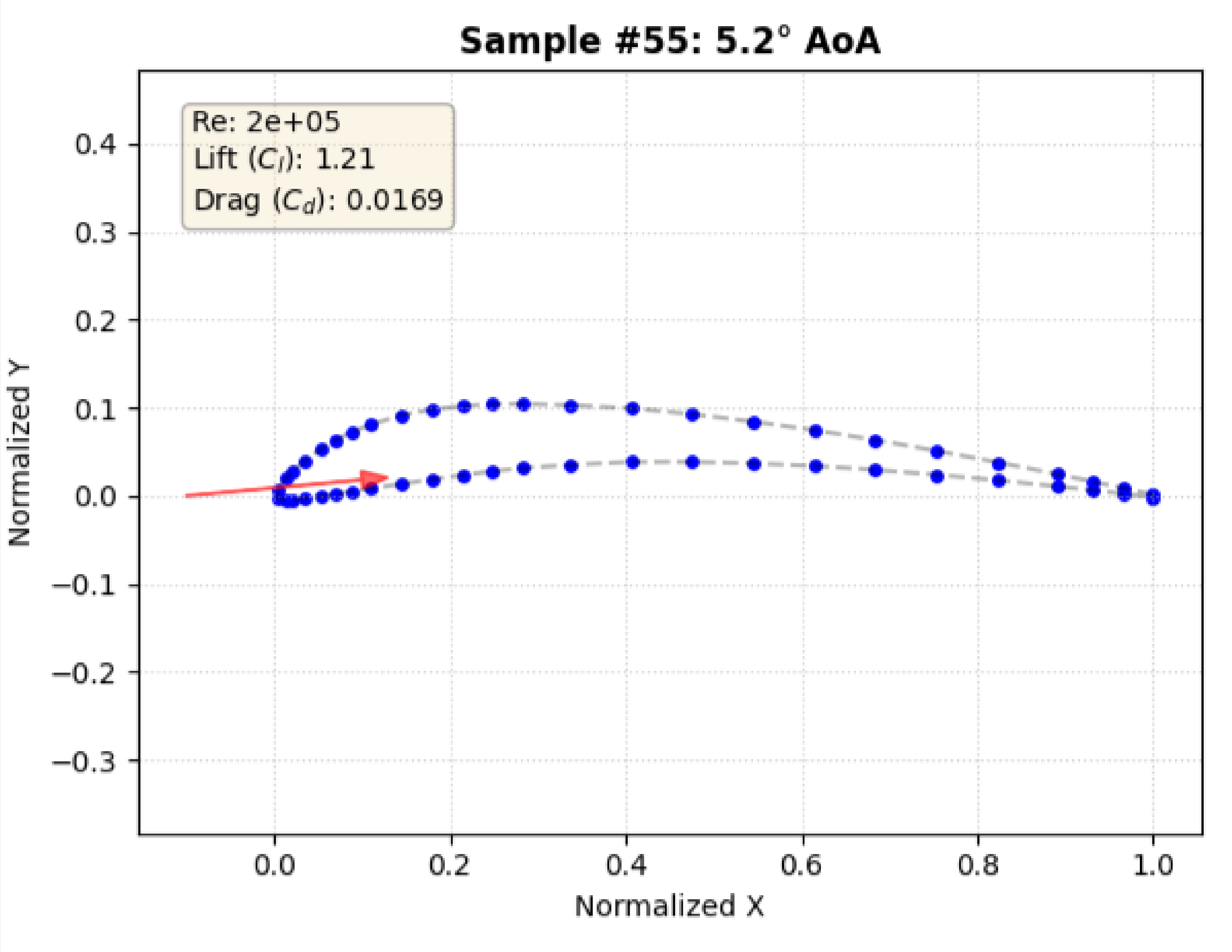
- CFD is Slow takes minutes or hours per simulations.
- Bottleneck: Testing 1000s of designs takes days.

SOLUTION

- Deep learning is fast, takes milliseconds per prediction.
- Approximation: Learns the mapping : Geometry to Forces

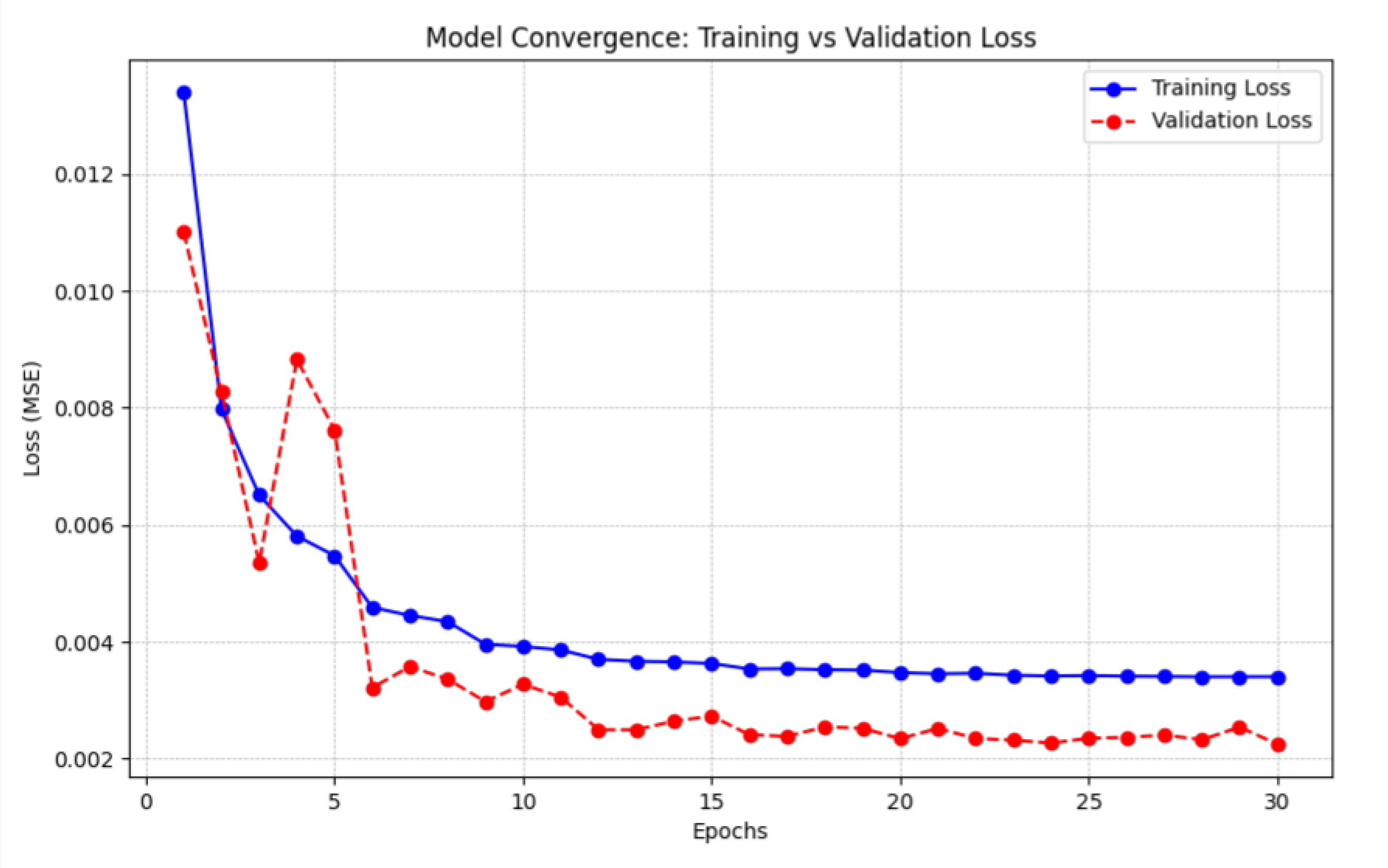
Dataset and Pipeline design

- **Dataset:** *DeepLearWing* (Kaggle) – CFD Data for thousands of airfoils.
- **Critical Preprocessing:** Raw data had inconsistent point counts.
- **Solution:** Resampling algorithms to fix all inputs to 50 coordinates.
- Model Input: 102 Features(100 Geometry + AOA + Reynolds)
- Model Structure: MLP Funnel (102->256 ->128->64->2).



Training Strategy

- **The Issue:** Drag Coefficient is tiny (0.001 - 2) and highly nonlinear.
- **Initial Result:** Model stagnated at $R^2 = 0.79$.
- **Solution:** Dynamic Learning Rate Schedular.
- Monitors the validation loss.
- Reduces LR when progress stalls.



Deploy

Applied Deep Learning: Airfoil Surrogate Model

1. Geometry Input

Source:

- (•) Dataset Sample
- ( ) Upload CSV (Unseen Data)

Sample ID

0 − +

Loaded Sample #0

2. Flow Conditions

Angle of Attack (°)

0.00

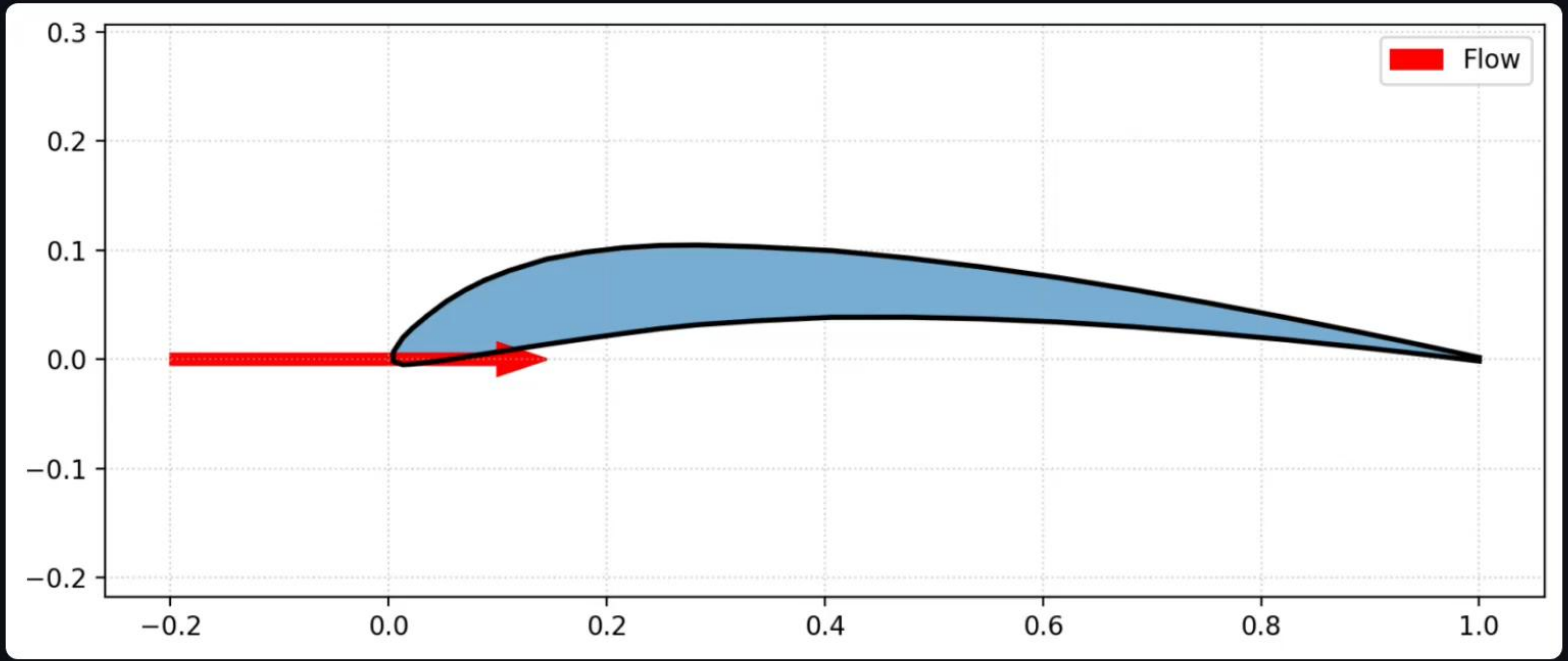
0,00

Reynolds Number

290000

290000

Geometry Visualization



Neural Network Prediction

Lift (C_l)

0.6144

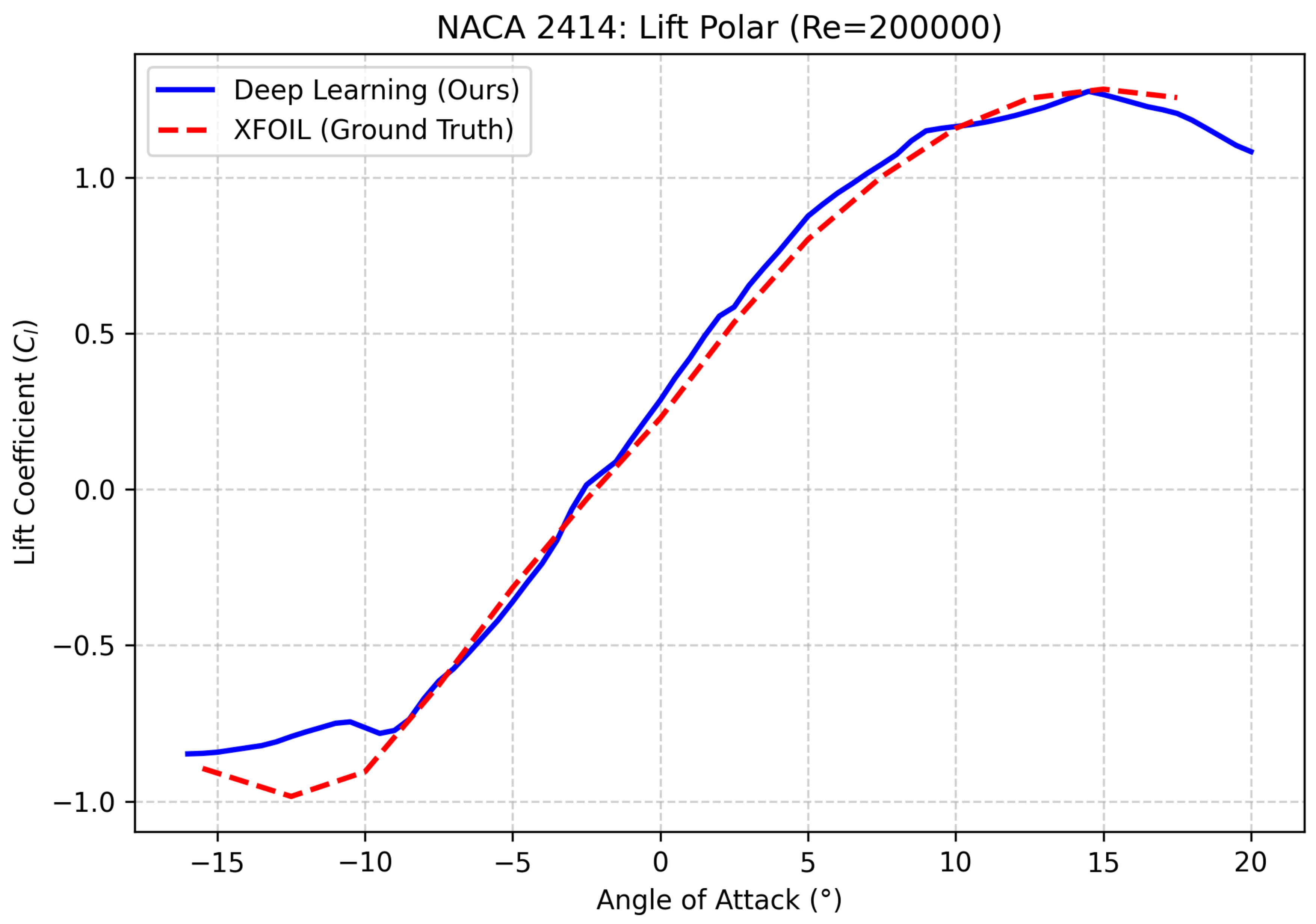
Drag (C_d)

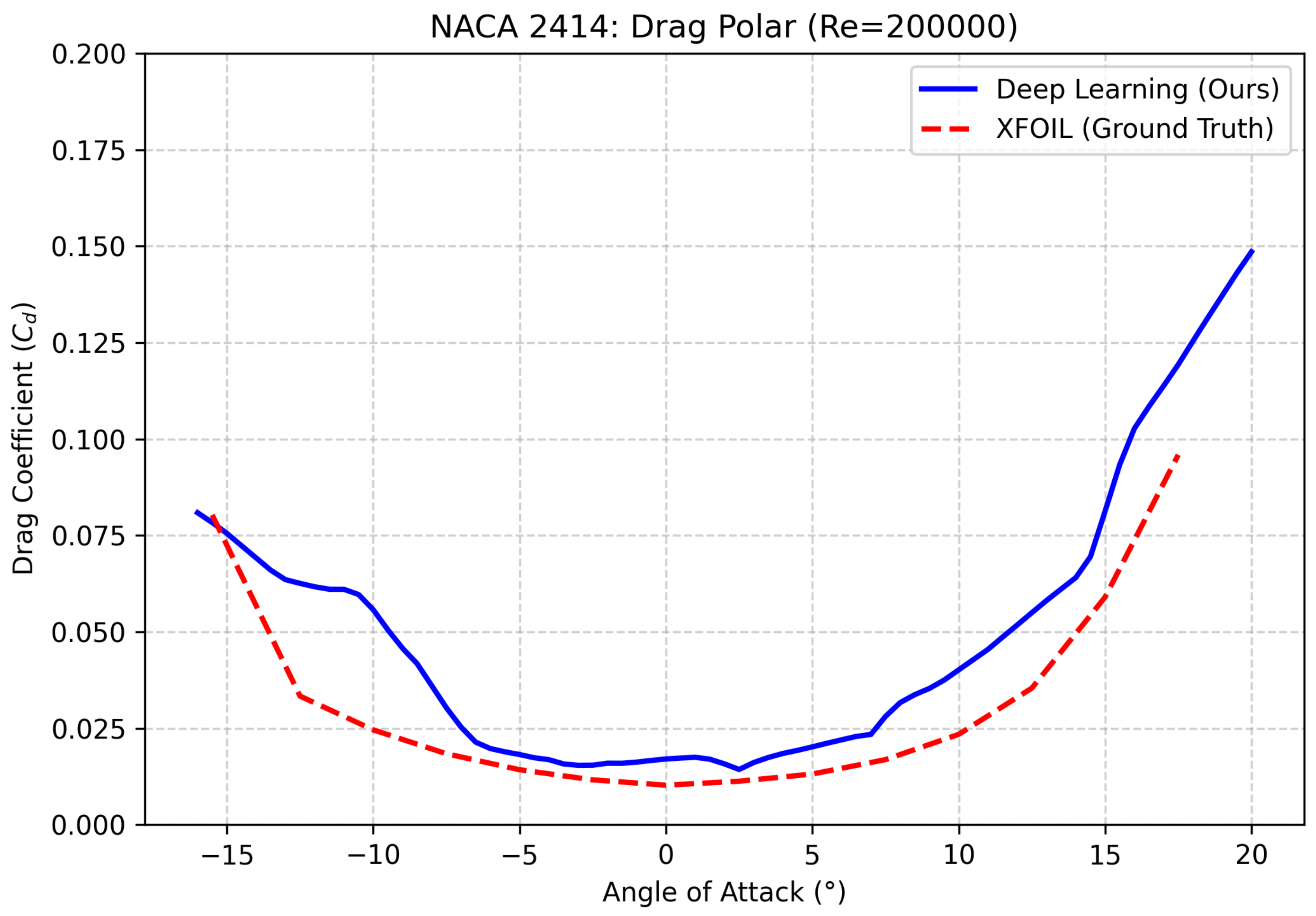
0.01063

Inference Time: 0.001461 sec

Validation on unseen Airfoils

NACA 2414 Airfoil(not in training set)Neural network vs XFOIL Analysis.





Key insights and future Scope

- Data Quality: Resampling was more effective than adding layers.
- Schedular was critical for drag convergence.
- Future Scopes: Capturing spatial curvature features with 1D-CNNs better than MLPs.
- Another Method: Predicting geometry from coefficients(GANs/VAEs)

THANK YOU